

Natural Capital and the Measurement of Productivity

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Wednesday 5th August, 2026

This paper studies how to measure economic growth after accounting for what is explained by investment and demographics.

- Productivity growth is an important indicator
 - Budget forecasts, cross-country comparisons, policy evaluation ...
- Standard (growth-accounting) measures do not account for environmental inputs
 - Examples: ecosystem services, subsurface resources, climate, ...
- Omitting inputs biases estimates of total factor productivity (TFP)
 - When natural capital (NK) enters production but is excluded from TFP measurement, the resulting residual contains a form of omitted-variable bias.

Motivation II

Expanding standard growth-accounting practices to account for the environment faces several difficulties:

- ① Income generated by natural capital typically goes unmeasured
 - Difficulties include public goods, apportionment, and accounting boundaries
 - Growth accounting requires measured market income and quantity changes
- ② Identifying and measuring individual service flows is difficult, even when stocks are observed
 - TFP accounting requires changes in service flows, not only changes in capital stocks
 - Stocks are easier to measure than the flows they provide
- ③ Better measurement *does not necessarily* imply more accurate productivity growth
 - More variables $\neq$ lower bias, even with perfect measurement

Research Question

We focus on the third measurement challenge.

Research question

When does measuring natural capital improve estimates of productivity growth?

Approach this question in three steps:

- 1 Relevance: Is there robust evidence that NK affects output?
- 2 Theory: Under what conditions does additional measurement improve estimates of productivity growth?
- 3 Empirics: Are existing data consistent with these conditions?

- 1 Show existing estimates of TFP growth are highly correlated with benchmark natural capital stocks $\implies$ omitting NK likely biases measured TFP.
- 2 Derive conditions sharp conditions when additional measurement reduces measurement error for residual-based estimates of productivity growth.
- 3 Give simulation evidence that accounting for non-renewable NK use in existing measures of TFP induces policy-relevant reductions in measurement error for most countries: 7 bps at the median or 13 percent of average historical annual TFP growth in sample.

Modeling Economic Growth

We consider a neoclassical economy with $N > 3$ factor inputs. Production is

$$Y(t) = TFP(t) \prod_{x \in \mathbf{X}} X_x(t)^{\gamma_x(t)} \quad (1)$$

- $\mathbf{X}$ – Set of N factor inputs
- $\gamma_x(t)$ – Output elasticities (factor shares under CRS with competitive factor markets)
- $TFP(t)$ – *Total Factor Productivity* (Solow, 1956)

TFP Growth in Theory

Taking logs and differentiating (1) with respect to time, output growth is

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \mathbf{X}} \gamma_x(t) g_x(t) \quad (2)$$

Equation (2) is the growth-accounting equation: it maps changes in factor inputs to changes in output. When output elasticities are autonomous

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \mathbf{X}} \gamma_x g_x(t) \quad (3)$$

TFP growth is, *by definition*, output growth that is unexplained by input growth. In this setting, the residual coincides with the unobservable theoretical object in (2).

Measurement in Practice

TFP growth is unobservable as true production function is unknown. The Solow residual (Solow 1957) estimates TFP growth as the difference between output growth and growth in *included* inputs I .

$$g_A \triangleq g_Y - \sum_{i \in I} \gamma_i g_i \quad (4)$$

In practice (e.g., Penn World Tables, World Bank, BLS, Eurostat, ...), implemented as

$$g_A = g_Y - \alpha G^K(\{g_k\}) - (1 - \alpha) G^L(\{g_l\}) \quad (5)$$

where G^K and G^L measure growth in indices of produced capital and labor, and α is the estimated capital share. NK is not usually considered.

Testable Predictions

This residual-based estimator can be decomposed into two additional terms in addition to underlying TFP growth, g_{TFP} :

$$g_A(t)|I, \hat{\gamma}, \hat{\mathbf{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x \in \{\mathbf{X} \setminus I\}} \gamma_x g_x(t)}_{\text{Omitted inputs}} + \underbrace{\sum_{i \in I} [\gamma_i g_i(t) - \hat{\gamma}_i \hat{g}_i(t)]}_{\text{Measurement error}}. \quad (6)$$

A lack of correlation between $g_A(t)$ and changes in services from natural capital is (1) testable and (2) partially falsifies that these services are omitted inputs to production (i.e., matter). Reduced form approach will try and falsify this correlation.

Estimating Equation

A form of correlation between NK and measured TFP growth is testable. We estimate (7) using a balanced panel of 117 countries. The outcome is TFP growth from the Penn World Table (Feenstra et al. 2015):

$$g_{A_{j,t}} = \sum_{s \in \mathbf{S}} \beta_s g_{S_{s,j,t}} + \theta' \mathbf{Z}_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t}, \quad (7)$$

- $\mathbf{S}$: six natural capital stocks from the World Bank's *Changing Wealth of Nations* project
- β_s : estimands of interest
- $\mathbf{Z}_{j,t}$: controls
- (δ_j, ζ_t) : country and year fixed effects

Identification and Inference I

Joint null: no correlation between changes in NK stocks and TFP

$$H_0 : \beta = \mathbf{0}. \quad (8)$$

However, important caveats to interpretation:

- ① **S** are stocks, not flows
 - Simple analogy: labor force vs. hours worked.
- ② Changes in stocks are *instruments* for omitted NK inputs: the β estimands identify

$$\beta_s = \sum_{x \in \mathbf{X} \setminus I} \gamma_x \frac{\text{Cov}(g_{x,j,t}, \tilde{g}_{s_{j,t}})}{\text{Var}(\tilde{g}_{s_{j,t}})}, \quad (9)$$

where relevance gives $\text{Cov}(\cdot, \cdot) \neq 0$ and omission implies $\gamma_x > 0$

- ③ Rejecting null then (partially) falsifies that instrumented services have $\gamma_x > 0$

Identification and Inference II

Other complications, namely the components of the ► Solow residual can lead to false positives:

- 1 NK growth is correlated with measurement error terms
- 2 NK growth is correlated with true TFP growth

even when the services that are instrumented do not affect output. False negatives can also arise if

- 3 Measurement error and correlation with true TFP growth offset correlations with omitted inputs
- 4 Covariances in (9) are not all of the same sign

Natural Capital S from the CWON

Resource	Natural unit	Associated service flows	Potential confounders
Timber land	Hectares	Timber harvest; carbon capture	Declining seeding services
Agricultural land	Square kilometers	Grazing area; soil services	Declining ecosystem services from fallow land
Non-timber forest cover	Square kilometers	Recreation; ecosystem services	Declining grazing area and forestry
Mangrove area	Hectares	Flood protection; water filtration; habitat provision	Marine recreation
Fishery biomass	Metric tons	Harvested fish biomass	Nutrient cycling; biological-pump dynamics
Hydroelectric generation	Gigawatt-hours	Electricity; flood prevention; water storage	River and floodplain ecosystem services

Note: Data are from the World Bank's *Changing Wealth of Nations* database (World Bank 2024). Associated service flows are examples of TFP-relevant services likely to be correlated with changes in each measured stock. Potential confounders include omitted productive inputs and measurement error in included factors.

TFP Growth vs. Growth in Renewable Natural Capital

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Timber land	0.165 (0.049)	0.168 (0.046)	0.071 (0.048)	0.074 (0.046)	-0.004 (0.053)	0.025 (0.104)
Agricultural land	0.029 (0.019)	0.045 (0.019)	0.032 (0.019)	0.038 (0.018)	0.042 (0.018)	0.043 (0.017)
Forest cover	-0.069 (0.062)	-0.113 (0.065)	-0.006 (0.062)	0.010 (0.062)	0.081 (0.055)	-0.027 (0.076)
Mangroves	0.112 (0.129)	0.057 (0.132)	0.352 (0.163)	0.322 (0.167)	0.318 (0.202)	-0.111 (0.427)
Fishery biomass	0.112 (0.056)	0.026 (0.054)	0.092 (0.063)	0.070 (0.061)	0.039 (0.082)	-0.109 (0.105)
Hydropower	0.010 (0.005)	0.011 (0.005)	0.009 (0.005)	0.009 (0.005)	0.006 (0.006)	0.006 (0.003)
<i>Specification</i>						
Factor input controls	No	Yes	No	Yes	Yes	Yes
Country and year FE	No	No	Yes	Yes	Yes	Yes
Country trends	No	No	No	No	Yes	No
Outcome lags	No	No	No	No	No	Yes
Wald p-value	0.001	0.001	0.018	0.018	0.090	0.045
Bootstrap Wald p-value	0.000	0.000	0.008	0.010	-	0.062
Observations	2,808	2,808	2,808	2,808	2,808	2,457
Countries	117	117	117	117	117	117

▶ Alt. Covariate Selection

▶ Exclude Land

▶ Alternative TFP Series

▶ RI Test

Thought Experiment: When *Would* Measurement Help?

Suppose that two natural-capital assets contribute to production alongside the standard neoclassical factors. Output is

$$Y_t = TFP_t K_t^{\gamma_K} L_t^{\gamma_L} N_{1t}^{\gamma_1} N_{2t}^{\gamma_2} \quad (10)$$

- K – produced capital
- L – labor
- N_1 – nonrenewable capital services
- N_2 – renewable capital services

The correctly specified expression for TFP growth is

$$g_{TFP}(t) = g_Y(t) - \gamma_K g_K(t) - \gamma_L g_L(t) - \gamma_1 g_{N_1}(t) - \gamma_2 g_{N_2}(t) \quad (11)$$

Estimating TFP with Incomplete Information

A national accountant wants to estimate historical TFP growth for this economy. Assyme Suspects it may be important, but cannot measure N_2 (renewables). Instead must choose between two residual-based estimators of TFP: one that includes only produced capital and labor

$$\hat{g}_A \triangleq \bar{g}_Y - \hat{\gamma}_K \bar{g}_K - \hat{\gamma}_L \bar{g}_L \quad (\text{Model 1})$$

and a second that also includes nonrenewables.

$$\tilde{g}_A \triangleq \bar{g}_Y - \tilde{\gamma}_K \bar{g}_K - \tilde{\gamma}_L \bar{g}_L - \tilde{\gamma}_1 \bar{g}_{N_1} \quad (\text{Model 2})$$

Assume her goal is to choose the estimator with the lowest RMSE over a 25-year sample. **Question 1:** does going from model 1 to model 2 help?

Even if the national accountant knows g_Y, g_K, g_L, g_{N_1} and their associated output elasticities without error, it is ambiguous whether adding NK—that is, moving from Model 1 to Model 2—reduces the bias or RMSE of estimated TFP (Proposition 1).

The improvement is unambiguous only when either (i) the number of NK assets is large or (ii) the variance in their contributions to growth is small (Proposition 2).

The realized improvement from implementing Models 1 and 2 depends on the underlying DGP for output. **Question 2:** does going from model 1 to model 2 help given what we've observed

Calibrated thought Experiment

Calculating what the change in RMSE *would be* requires calibrating a baseline economy. We calibrate the following moments for 100 countries:

Table: Simulation Moments and Sources

Population Moment	Sample Moment	Source
$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON
$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON
$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Σ	$\hat{\Sigma}$	CWON & PWT

Neoclassical moments equal to sample analogs in PWT. Growth in non-renewables and renewables more complicated, see paper.

Estimating γ s

We assume the national accountant observes time series of $\{g_Y, g_K, g_L, g_{N_1}\}$ without error. She uses OLS to estimate the output elasticities for each model

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \quad (\text{Model 1})$$

and

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \quad (\text{Model 2})$$

The intercept terms $\hat{\gamma}_A$ and $\tilde{\gamma}_A$, estimands of average TFP growth, are objects of interest.

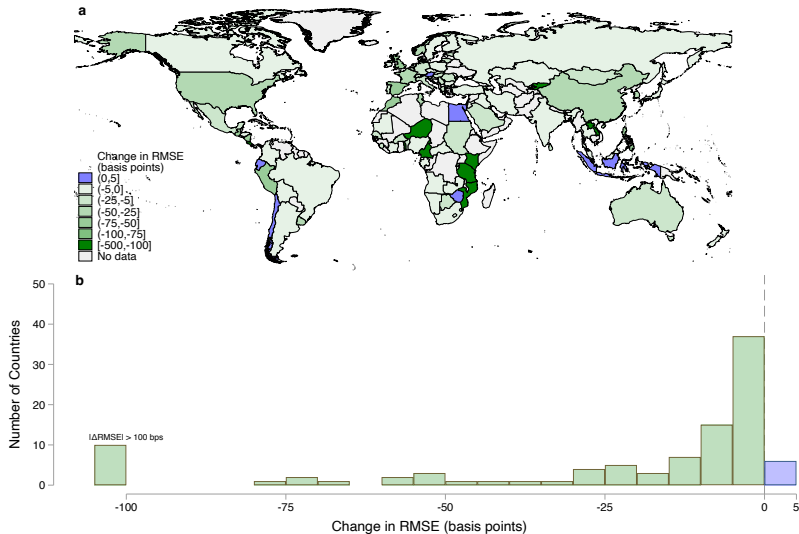
Performance metric: the reduction in expected RMSE going from model 1 to model 2

$$\Delta \text{RMSE} | \mathbf{g} = \sqrt{\text{Var}(\hat{\gamma}_A | \hat{\mathbf{g}}) + \mathbb{E}[\varepsilon | \hat{\mathbf{g}}]^2} - \sqrt{\text{Var}(\tilde{\gamma}_A | \tilde{\mathbf{g}}) + \mathbb{E}[\delta | \tilde{\mathbf{g}}]^2} \quad (12)$$

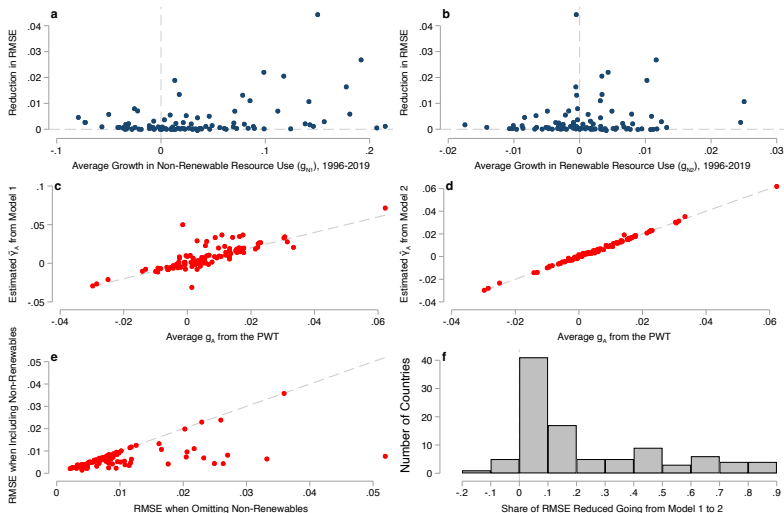
where δ and ε are conditional biases.

Assume the national accountant prefers the model with lower expected RMSE.

Distribution of RMSE Reduction



Cross-Sectional Results



Conclusion

- 1 Omitting environmental inputs induces systematic TFP bias and estimation error.
- 2 Including omitted variables is not guaranteed to improve estimates.
- 3 There is strong empirical evidence that this bias is present, although it is difficult to rule out other explanations.
- 4 Calibrated simulations suggest sizable bias reductions for most countries over the past 25 years.

Thanks

Thanks for listening. Questions, comments, and feedback are welcome.

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Robustness: Systematic Covariate Selection

Model	Estimator	p (Wald)	p (Wild bootstrap)
BCH (2014)	PDS	0.002	0.004
BCH (2014)	LASSO	0.000	0.007
BCH (2014)	post-LASSO	0.001	0.007
D-K (1998) SEs	TWFE	0.000	0.000
D-K (1998) SEs	TWFE/Controls	0.000	0.000
D-K (1998) SEs	TWFE/Controls/Trends	0.000	0.004

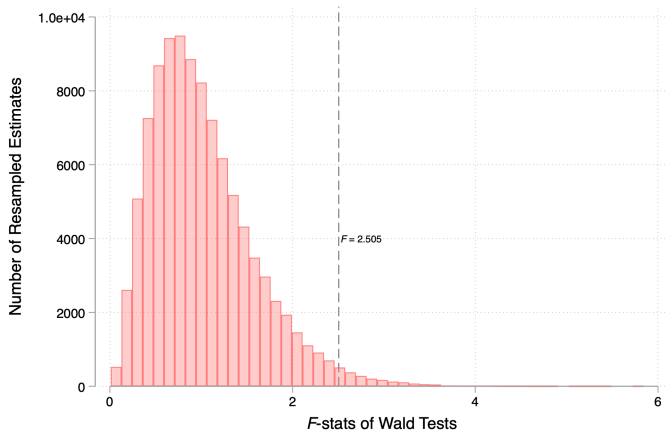
[◀ Back to Main Results](#)

Robustness: Excluding Land

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Forest cover	0.068 (0.061)	0.027 (0.062)	0.047 (0.068)	0.067 (0.075)	0.086 (0.043)	-0.013 (0.070)
Mangroves	0.138 (0.240)	0.085 (0.237)	0.348 (0.298)	0.317 (0.322)	0.303 (0.366)	-0.096 (0.428)
Biomass	0.114 (0.098)	0.031 (0.078)	0.092 (0.107)	0.070 (0.101)	0.039 (0.134)	-0.109 (0.107)
Hydropower	0.011 (0.004)	0.011 (0.004)	0.009 (0.004)	0.009 (0.004)	0.006 (0.003)	0.006 (0.003)
<i>Specification</i>						
Factor input controls	No	Yes	No	Yes	Yes	Yes
Country and year FE	No	No	Yes	Yes	Yes	Yes
Country trends	No	No	No	No	Yes	No
Outcome lags	No	No	No	No	No	Yes
Wald p-value	0.050	0.061	0.111	0.107	0.062	0.229
Bootstrap Wald p-value	0.067	0.108	0.128	0.116	-	0.185
Observations	2,808	2,808	2,808	2,808	2,808	2,457

[← Back to Main Results](#)

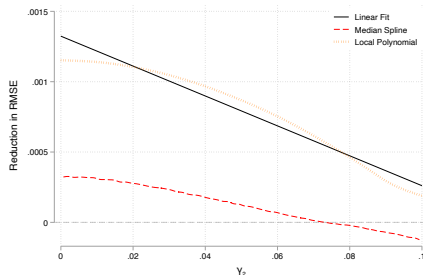
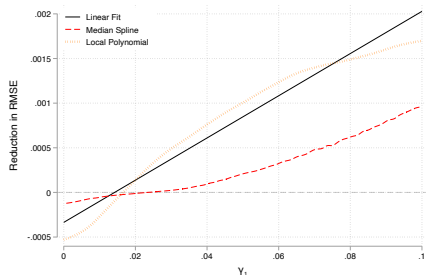
Robustness: Randomization-Inference Test



Nonparametric test: $p = 0.0185$.

[◀ Back to Main Results](#)

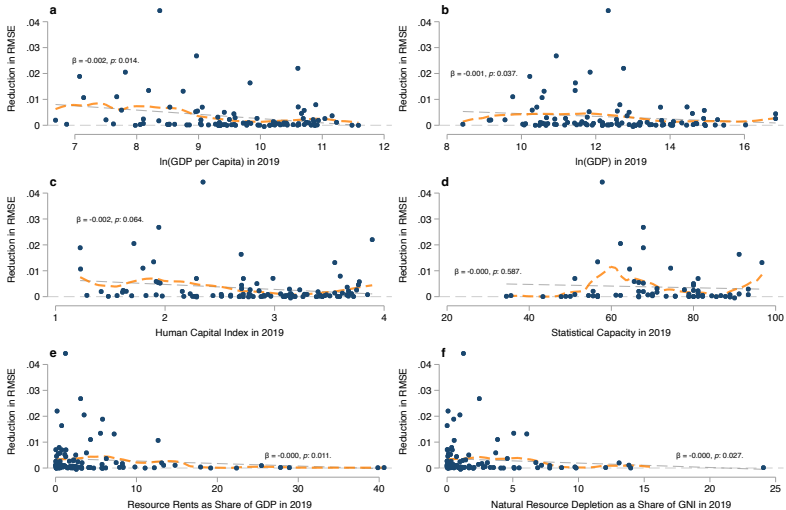
Sensitivity of RMSE Reductions to Calibration



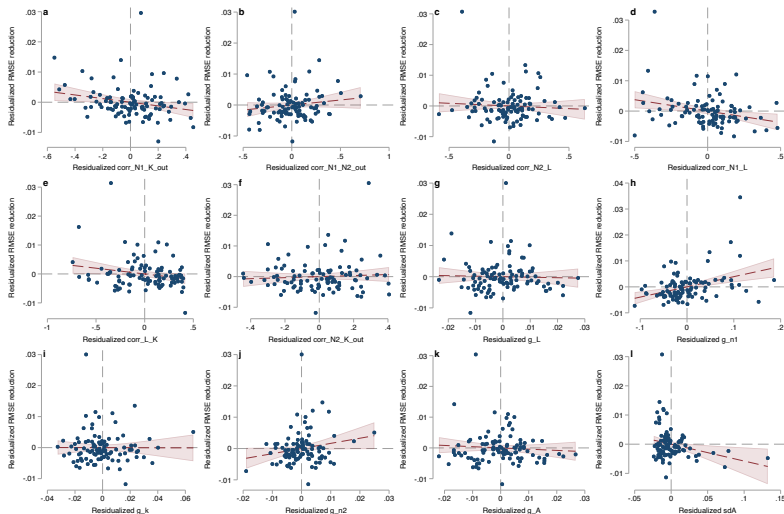
(a) Sensitivity of ΔRMSE to Calibrated γ_1 (b) Sensitivity of ΔRMSE to Calibrated γ_2

Figure: Linear, median-spline, and local-polynomial fits relating country-level ΔRMSE_j to draws of γ_1 and γ_2 , based on $N = 10,000,000$ Monte Carlo simulations.

RMSE Reductions and Other Determinants of Wealth



Cross-Sections of RMSE Reductions



[◀ Back to Results](#)